MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BBA (HONOURS) REGULAR EXAMINATION MARCH 2026
FOURTH SEMESTER Core Course(CC) - (2024 ADMISSION ONWARDS)
FINANCIAL MANAGEMENT - MG4CCRBA201- ANSWERKEY

Part A (Short Answer Questions (3 Marks Each))

1a. Write any three characteristics of debentures?

- Creditorship Security: Debenture holders are creditors of the company, not owners.
- Fixed Interest: They carry a fixed rate of interest payable regardless of profit levels.
- Maturity: They are usually issued for a specific period and are redeemable at the end of that term.

1b. Define Financial Management:

Financial management is the activity concerned with the planning, raising, controlling, and administering of funds used in the business. It involves making three key decisions: Investment (where to put money), Financing (where to get money), and Dividend (how much to return to owners). The primary goal is usually the maximization of shareholder wealth.

2a. Define Capitalisation:

Capitalisation refers to the total amount of long-term funds (equity, reserves, and debt) available in a business. It represents the par value of the company's stocks and bonds.

2b. What are the assumptions of Net Operating Income (NOI) Approach?

- The market capitalizes the value of the firm as a whole.
- The overall cost of capital remains constant.
- The cost of debt remains constant.
- There are no corporate taxes.

3a. What is Composite Leverage?

Composite leverage (also known as Combined Leverage) expresses the relationship between a change in Sales and the corresponding change in Taxable Income (EPS). It measures the total risk of the firm by combining both Operating Leverage (fixed costs) and Financial Leverage (fixed interest charges). Combined Leverage = DOL * DFL

3b. Define Gross Working Capital:

Gross Working Capital refers to the total investment of a company in its current assets (Cash, Inventory, Receivables, etc.).

4a. Two forms of dividend payments:

- Cash Dividend: Paid in liquid cash to shareholders.
- Stock Dividend (Bonus Shares): Paid in the form of additional shares.

4b. What is Payback Period method?

The Payback Period is the length of time required to recover the initial cost of an investment from its net cash inflows. It is a simple capital budgeting technique where a shorter payback period is generally preferred as it indicates lower risk and faster liquidity.

5a. Three advantages of Retained Earnings:

- Cheapest source: No flotation costs or legal formalities.
- Financial Independence: No dependence on external creditors.
- Tax Benefits: Shareholders save on personal dividend taxes.

5b. Distinguish between Profit maximization and wealth maximization?

Profit Maximisation

- Time Horizon: It is a short-term objective.
- Primary Aim: It aims at maximising the profit of the firm.
- Measurement: It measures the effectiveness of the organisation.
- Time Value: It does not consider the time value of money.
- Earnings Strategy**:** It provides no clarity on whether the earnings would be distributed or retained in the firm.

Wealth Maximisation

- Time Horizon: It is a long-term objective.
- Primary Aim: It aims at maximising the wealth of the shareholders.
- Measurement: It measures the financial stability of the organisation.
- Time Value: It considers the time value of money.
- Earnings Strategy: It distributes the earnings among shareholders as dividends.

Part B (Short Essay Questions (5 Marks Each)

6a. Importance of Investment, Financing, and Dividend Decisions:

- Investment Decision: Determines which assets to buy to ensure maximum return (e.g., Capital Budgeting).
- Financing Decision: Determines the best mix of debt and equity to minimize the cost of capital.
- Dividend Decision: Balances the need to reward shareholders with the need to retain funds for growth.

6b. Compare Equity vs. Preference Shares:

Shares

- Capital Type: Shares represent a part of the owned capital of the company.
- Status of Holder: Shareholders are considered the owners of the company.
- Security: Shares do not have any charge on the assets of the company; they are generally unsecured.
- Return on Investment: The dividend given to shareholders may fluctuate from year to year based on the amount of profit earned.
- Payment Condition: Dividends on shares are paid only when the company makes a profit.

- Voting Rights: Shareholders, as members of the company, enjoy full voting rights.

Debentures

- Capital Type: Debentures constitute a loan provided to the company.
- Status of Holder: Debenture holders are creditors of the company.
- Security: Debentures usually have a charge on the assets of the company, meaning they are secured.
- Return on Investment: Debentures carry a fixed rate of interest.
- Payment Condition: Interest on debentures must be paid regardless of whether the company makes a profit or not.
- Voting Rights: Debenture holders do not have any voting rights because they are creditors, not members.

7a. Compute Operating Cycle:

The operating cycle is the total time it takes for a company to convert its cash into inventory and then back into cash from sales.

- Formula: OC = R + W + F + D – C
    - R (Raw Material Period) = 40 days
    - W (WIP Period) = 20 days
    - F (Finished Goods Period) = 20 days
    - D (Debtors Period) = 30 days
    - C (Creditors Period) = 20 days

Calculation: 40 + 20 + 20 + 30 – 20 = 90days

7b. Evaluate the suitability of Just-in -time system in reducing working capital

Just-in-Time (JIT) reduces working capital by ensuring inventory arrives only when needed.

- Pros: Lowers storage costs, reduces wastage, and frees up cash tied in stock.
- Cons: Requires highly reliable suppliers and efficient logistics. Any delay in the supply chain can halt production.

How JIT Reduces Working Capital

- Working capital is essentially the liquidity required for day-to-day operations, often tied up in current assets. JIT targets several key components:
- Reduction in Inventory Investment: Since materials arrive only when a production order is placed, the “Raw Materials” component of working capital is minimized. Similarly, “Finished Goods” inventory is kept low as items are shipped immediately upon completion.
- Lower Holding Costs: By eliminating large stockpiles, a company reduces costs associated with warehousing, insurance, security, and physical deterioration. This improves the Operating Cash Flow.
- Improved Inventory Turnover: JIT increases the frequency with which inventory is replaced, leading to a higher Inventory Turnover Ratio. This means cash is “locked” in the production cycle for a much shorter period.
- Minimal Obsolescence: In industries with fast-changing technology (like electronics), JIT prevents the firm from holding large amounts of stock that might become worthless or “dead capital.”

8a. Role of Time Value of Money (TVM) in Capital Budgeting:

TVM recognizes that a rupee today is worth more than a rupee tomorrow due to inflation, risk, and opportunity cost. Methods like NPV and IRR use TVM to discount future cash flows, ensuring more accurate investment decisions.

TVM-Based Capital Budgeting Techniques
Financial managers use TVM to calculate the "True Value" of a project through these primary methods:

e) Net Present Value (NPV)

This is the gold standard. It converts all future cash inflows into today's value and subtracts the initial investment.

- Rule: If NPV > 0, the project adds wealth to the firm.

B. Internal Rate of Return (IRR)

The IRR is the specific discount rate that makes the NPV of a project equal to zero. It represents the project's expected annual rate of return.

- Rule: If IRR > Cost of Capital, the project is accepted.

C. Discounted Payback Period

Unlike the simple payback period (which just adds up raw cash), the discounted version discounts each cash flow first. This tells the manager exactly how long it takes to recover the investment in "today's terms."

8b. Payback Period Calculation:

Initial Investment = ₹1,00,000.

- Project A: Cumulative Cash Flow reaches 1,00,000 during Year 4.
  Payback = 3 years + (25,000 / 40,000) = 3.625 years
- Project B: Cumulative Cash Flow reaches exactly 1,00,000 at the end of Year 4.
  Payback = 4years

Project A is better because it recovers the investment faster.

Part C (Long Essay (20 Marks Each)

9a. What is capital gearing? Explain the factors governing capital gearing?

Capital Gearing (also known as Financial Leverage) refers to the relationship between a company's fixed-interest-bearing funds (like debentures, bonds, and preference shares) and its equity-shareholder funds (equity share capital, reserves, and surplus).

Levels of Capital Gearing

- High Gearing: Occurs when the proportion of fixed-cost capital (Debt/Preference Shares) is high relative to equity capital. This is aggressive and increases financial risk but can lead to higher EPS if the company is profitable.
- Low Gearing: Occurs when the proportion of equity capital is significantly higher than debt. This is a conservative approach with lower financial risk.

Factors Governing Capital Gearing

A financial manager must consider several internal and external factors before deciding on the gear ratio:- Nature of Business, Cost of Choice of Finance, Growth and Expansion Phase, Control and Voting Power, Capital Market Conditions, Asset Structure

9b. A Company has a share capital of ₹2,00,000 divided in to shares of ₹10 each. It has a major expansion project and needs to raise additional capital of another ₹1,00,000. As a financial analyst, evaluate the alternative financing options available to the company and analyse their impact on shareholders' earnings. The objective is to apply financial management concepts to identify the most suitable source of finance by examining its effect on Earnings Per Share (EPS) under different EBIT conditions. Alternative Options: A) Issue of 10,000 equity shares of ₹10 each B) Issue of 10,000, 12% preference shares of ₹10 each C) Issue of 10% Debentures of ₹1,00,000

The company's present Earnings before Interest and Tax (EBIT) are ₹80,000 PA. You are required to calculate the effect of each of the above mode of financing on earnings per share presuming: A) EBIT continues to be the same even after expansion B) EBIT increases by ₹20,000

Case 1: EBIT remains at ₹80,000

Particulars	**Option A (Equity)**	**Option B (Pref. Shares)**	**Option C (Debentures)**
EBIT	**₹80,000**	**₹80,000**	**₹80,000**
Less: Interest (10% of 1L)	**-**	**-**	**(₹10,000)**
EBT	**₹80,000**	**₹80,000**	**₹70,000**
Less: Tax @ 30%	**(₹24,000)**	**(₹24,000)**	**(₹21,000)**
EAT	**₹56,000**	**₹56,000**	**₹49,000**
Less: Pref. Dividend (12%)	**-**	**(₹12,000)**	**-**
Earnings for Equity	**₹56,000**	**₹44,000**	**₹49,000**
No. of Equity Shares	**30,000**	**20,000**	**20,000**

Particulars	Option A (Equity)	Option B (Pref. Shares)	Option C (Debentures)
EPS (Earnings / Shares)	₹1.87	₹2.20	₹2.45

Case 2: EBIT increases by ₹20,000 (Total EBIT = ₹1,00,000)

Particulars	Option A (Equity)	Option B (Pref. Shares)	Option C (Debentures)
EBIT	₹1,00,000	₹1,00,000	₹1,00,000
Less: Interest	-	-	(₹10,000)
EBT	₹1,00,000	₹1,00,000	₹90,000
Less: Tax @ 30%	(₹30,000)	(₹30,000)	(₹27,000)
EAT	₹70,000	₹70,000	₹63,000
Less: Pref. Dividend	-	(₹12,000)	-
Earnings for Equity	₹70,000	₹58,000	₹63,000
No. of Equity Shares	30,000	20,000	20,000
EPS (Earnings / Shares)	₹2.33	₹2.90	₹3.15

Conclusion: In both scenarios, Option C (Debentures) provides the highest EPS. This is because debt is a cheaper source of finance due to the tax-deductibility of interest, creating positive financial leverage.

10a. What is WACC? How it is determined? State the advantages and disadvantages of computing cost of capital using WACC?

WEIGHTED AVERAGE COST OF CAPITAL

Weighted average cost of capital is the average cost of the costs of various sources of financing. It is also known as composite cost of capital or overall cost of capital. The computation of Weighted Average cost of capital is done by putting weights to the specific cost of capital in proportion of the various sources of funds to the total. The weights may be given either by using the book value or market value of the source. Weighted Average cost of capital can be computed as follows:
Kw = sum WX\sum W
Where:

- X = Cost of specific source of finance
- W = Weight, proportion of specific source of finance

The computation of overall cost of capital involves the following steps:

- a. Assigning weights to specific cost.
- b. Multiplying the cost of each of the sources by the appropriate weights.
- c. Dividing the total weighted cost by the total weight.

10b.

Particulars	Firm G (₹)	Firm H (₹)
Sales	50,00,000	50,00,000
Variable Cost	30,00,000	30,00,000
Fixed Cost	12,00,000	8,00,000
Interest	3,00,000	6,00,000

Questions:

- a) Calculate Contribution and EBIT
- b) Compute DOL, DFL and DCL
- c) Compare Operating Risk
- d) Compare Financial Risk
- e) Identify which firm is more risky overall with justification

Solution

a) Calculation of Contribution and EBIT

Particulars	Firm G (₹)	Firm H (₹)
Sales	50,00,000	50,00,000
Less: Variable Cost	(30,00,000)	(30,00,000)

Particulars	Firm G (₹)	Firm H (₹)
(i) Contribution	**20,00,000**	**20,00,000**
Less: Fixed Cost	(12,00,000)	(8,00,000)
(ii) EBIT	**8,00,000**	**12,00,000**
Less: Interest	(3,00,000)	(6,00,000)
EBT (Earnings Before Tax)	**5,00,000**	**6,00,000**

1. Degree of Operating Leverage (DOL) = Contribution/EBIT
    - Firm G: 20,00,000 / 8,00,000 = 2.5
    - Firm H: 20,00,000 / 12,00,000 = 1.67
2. Degree of Financial Leverage (DFL) = EBIT/EBT
    - Firm G: 8,00,000 / 5,00,000 = 1.6
    - Firm H: 12,00,000 / 6,00,000 = 2.0
3. Degree of Combined Leverage (DCL) = DOL*DFL
    - Firm G: 2.5 * 1.6 = 4
    - Firm H: 1.67 *2.0 = 3.34

g)Compare Operating Risk

- Firm G (2.5) has a higher DOL than Firm H (1.67).
- Conclusion: Firm G carries higher operating risk due to higher fixed operating costs.

d) Compare Financial Risk

Financial risk is measured by DFL.

- Firm H (2.0) has a higher DFL than Firm G (1.6).
- Conclusion: Firm H carries higher financial risk due to its higher interest burden.

e) Overall Risk and Justification

Overall risk is measured by DCL, which shows the total impact of both operating and financial fixed costs.

- Firm G (4.0) is more risky overall compared to Firm H (3.34).
- Justification: While Firm H has higher debt (financial risk), the significantly higher fixed operating costs in Firm G make it more sensitive to changes in sales. A 1% change in sales for Firm G will result in a 4% change in taxable income, whereas for Firm H, the change is only 3.34%.

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